High-Resolution Image Synthesis with Latent Diffusion Models

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https://github.com/CompVis/latent-diffusion

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Abstract

By decomposing the image formation process into a sequential application of denoising autoencoders, diffusion models (DMs) achieve state-of-the-art synthesis results on image data and beyond. Additionally, their formulation allows for a guiding mechanism to control the image generation process without retraining. However, since these models typically operate directly in pixel space, optimization of powerful DMs often consumes hundreds of GPU days and inference is expensive due to sequential evaluations. To enable DM training on limited computational resources while retaining their quality and flexibility, we apply them in the latent space of powerful pretrained autoencoders. In contrast to previous work, training diffusion models on such a representation allows for the first time to reach a near-optimal point between complexity reduction and detail preservation, greatly boosting visual fidelity. By introducing cross-attention layers into the model architecture, we turn diffusion models into powerful and flexible generators for general conditioning inputs such as text or bounding boxes and high-resolution synthesis becomes possible in a convolutional manner. Our latent diffusion models (LDMs) achieve new state-of-the-art scores for image inpainting and class-conditional image synthesis and highly competitive performance on various tasks, including text-to-image synthesis, unconditional image generation and super-resolution, while significantly reducing computational requirements compared to pixel-based DMs.

1. Introduction

Image synthesis is one of the computer vision fields with the most spectacular recent development, but also among those with the greatest computational demands. Especially high-resolution synthesis of complex, natural scenes is presently dominated by scaling up likelihood-based models, potentially containing billions of parameters in autoregressive (AR) transformers [66,67]. In contrast, the promising results of GANs [3, 27, 40] have been revealed to be mostly confined to data with comparably limited variability as their adversarial learning procedure does not easily scale to modeling complex, multi-modal distributions. Recently, diffusion models [82], which are built from a hierarchy of denoising autoencoders, have shown to achieve impressive

Figure 1. Boosting the upper bound on achievable quality with less agressive downsampling. Since diffusion models offer excellent inductive biases for spatial data, we do not need the heavy spatial downsampling of related generative models in latent space, but can still greatly reduce the dimensionality of the data via suitable autoencoding models, see Sec. 3. Images are from the DIV2K [1] validation set, evaluated at 512^2 px. We denote the spatial downsampling factor by f. Reconstruction FIDs [29] and PSNR are calculated on ImageNet-val. [12]; see also Tab. 8.

results in image synthesis [30,85] and beyond [7,45,48,57], and define the state-of-the-art in class-conditional image synthesis [15,31] and super-resolution [72]. Moreover, even unconditional DMs can readily be applied to tasks such as inpainting and colorization [85] or stroke-based synthesis [53], in contrast to other types of generative models [19,46,69]. Being likelihood-based models, they do not exhibit mode-collapse and training instabilities as GANs and, by heavily exploiting parameter sharing, they can model highly complex distributions of natural images without involving billions of parameters as in AR models [67].

Democratizing High-Resolution Image Synthesis DMs belong to the class of likelihood-based models, whose mode-covering behavior makes them prone to spend excessive amounts of capacity (and thus compute resources) on modeling imperceptible details of the data [16, 73]. Although the reweighted variational objective [30] aims to address this by undersampling the initial denoising steps, DMs are still computationally demanding, since training and evaluating such a model requires repeated function evaluations (and gradient computations) in the high-dimensional space of RGB images. As an example, training the most powerful DMs often takes hundreds of GPU days (*e.g.* 150 - 1000 V100 days in [15]) and repeated evaluations on a noisy version of the input space render also inference expensive,

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so that producing 50k samples takes approximately 5 days [15] on a single A100 GPU. This has two consequences for the research community and users in general: Firstly, training such a model requires massive computational resources only available to a small fraction of the field, and leaves a huge carbon footprint [65, 86]. Secondly, evaluating an already trained model is also expensive in time and memory, since the same model architecture must run sequentially for a large number of steps (*e.g.* 25 - 1000 steps in [15]).

To increase the accessibility of this powerful model class and at the same time reduce its significant resource consumption, a method is needed that reduces the computational complexity for both training and sampling. Reducing the computational demands of DMs without impairing their performance is, therefore, key to enhance their accessibility.

Departure to Latent Space Our approach starts with the analysis of already trained diffusion models in pixel space: Fig. 2 shows the rate-distortion trade-off of a trained model. As with any likelihood-based model, learning can be roughly divided into two stages: First is a *perceptual compression* stage which removes high-frequency details but still learns little semantic variation. In the second stage, the actual generative model learns the semantic and conceptual composition of the data (*semantic compression*). We thus aim to first find a *perceptually equivalent, but computationally more suitable space*, in which we will train diffusion models for high-resolution image synthesis.

Following common practice [11, 23, 66, 67, 96], we separate training into two distinct phases: First, we train an autoencoder which provides a lower-dimensional (and thereby efficient) representational space which is perceptually equivalent to the data space. Importantly, and in contrast to previous work [23, 66], we do not need to rely on excessive spatial compression, as we train DMs in the learned latent space, which exhibits better scaling properties with respect to the spatial dimensionality. The reduced complexity also provides efficient image generation from the latent space with a single network pass. We dub the resulting model class *Latent Diffusion Models* (LDMs).

A notable advantage of this approach is that we need to train the universal autoencoding stage only once and can therefore reuse it for multiple DM trainings or to explore possibly completely different tasks [81]. This enables efficient exploration of a large number of diffusion models for various image-to-image and text-to-image tasks. For the latter, we design an architecture that connects transformers to the DM's UNet backbone [71] and enables arbitrary types of token-based conditioning mechanisms, see Sec. 3.3.

In sum, our work makes the following **contributions**:

(i) In contrast to purely transformer-based approaches [23, 66], our method scales more graceful to higher dimensional data and can thus (a) work on a compression level which provides more faithful and detailed reconstructions than previous work (see Fig. 1) and (b) can be efficiently applied to high-resolution synthesis of megapixel images.

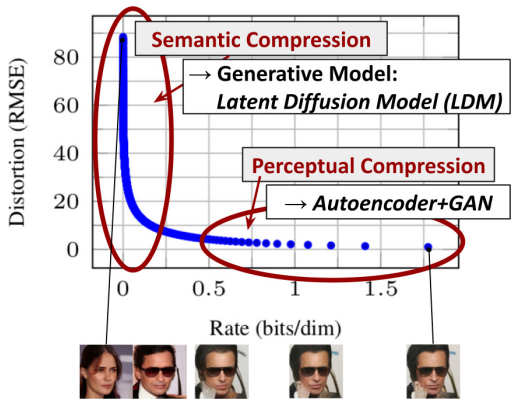

Figure 2. Illustrating perceptual and semantic compression: Most bits of a digital image correspond to imperceptible details. While DMs allow to suppress this semantically meaningless information by minimizing the responsible loss term, gradients (during training) and the neural network backbone (training and inference) still need to be evaluated on all pixels, leading to superfluous computations and unnecessarily expensive optimization and inference. We propose *latent diffusion models (LDMs)* as an effective generative model and a separate mild compression stage that only eliminates imperceptible details. Data and images from [30].

(ii) We achieve competitive performance on multiple tasks (unconditional image synthesis, inpainting, stochastic super-resolution) and datasets while significantly lowering computational costs. Compared to pixel-based diffusion approaches, we also significantly decrease inference costs.

(iii) We show that, in contrast to previous work [93] which learns both an encoder/decoder architecture and a score-based prior simultaneously, our approach does not require a delicate weighting of reconstruction and generative abilities. This ensures extremely faithful reconstructions and requires very little regularization of the latent space.

(iv) We find that for densely conditioned tasks such as super-resolution, inpainting and semantic synthesis, our model can be applied in a convolutional fashion and render large, consistent images of $\sim 1024^2$ px.

(v) Moreover, we design a general-purpose conditioning mechanism based on cross-attention, enabling multi-modal training. We use it to train class-conditional, text-to-image and layout-to-image models.

(vi) Finally, we release pretrained latent diffusion and autoencoding models at `https://github.com/CompVis/latent-diffusion` which might be reusable for a various tasks besides training of DMs [81].

2. Related Work

Generative Models for Image Synthesis The high dimensional nature of images presents distinct challenges to generative modeling. Generative Adversarial Networks (GAN) [27] allow for efficient sampling of high resolution images with good perceptual quality [3, 42], but are diffi-

cult to optimize [2, 28, 54] and struggle to capture the full data distribution [55]. In contrast, likelihood-based methods emphasize good density estimation which renders optimization more well-behaved. Variational autoencoders (VAE) [46] and flow-based models [18, 19] enable efficient synthesis of high resolution images [9, 44, 92], but sample quality is not on par with GANs. While autoregressive models (ARM) [6, 10, 94, 95] achieve strong performance in density estimation, computationally demanding architectures [97] and a sequential sampling process limit them to low resolution images. Because pixel based representations of images contain barely perceptible, high-frequency details [16,73], maximum-likelihood training spends a disproportionate amount of capacity on modeling them, resulting in long training times. To scale to higher resolutions, several two-stage approaches [23,67,101,103] use ARMs to model a compressed latent image space instead of raw pixels.

Recently, **Diffusion Probabilistic Models** (DM) [82], have achieved state-of-the-art results in density estimation [45] as well as in sample quality [15]. The generative power of these models stems from a natural fit to the inductive biases of image-like data when their underlying neural backbone is implemented as a UNet [15, 30, 71, 85]. The best synthesis quality is usually achieved when a reweighted objective [30] is used for training. In this case, the DM corresponds to a lossy compressor and allow to trade image quality for compression capabilities. Evaluating and optimizing these models in pixel space, however, has the downside of low inference speed and very high training costs. While the former can be partially adressed by advanced sampling strategies [47, 75, 84] and hierarchical approaches [31, 93], training on high-resolution image data always requires to calculate expensive gradients. We adress both drawbacks with our proposed *LDMs*, which work on a compressed latent space of lower dimensionality. This renders training computationally cheaper and speeds up inference with almost no reduction in synthesis quality (see Fig. 1).

Two-Stage Image Synthesis To mitigate the shortcomings of individual generative approaches, a lot of research [11, 23, 67, 70, 101, 103] has gone into combining the strengths of different methods into more efficient and performant models via a two stage approach. VQ-VAEs [67, 101] use autoregressive models to learn an expressive prior over a discretized latent space. [66] extend this approach to text-to-image generation by learning a joint distributation over discretized image and text representations. More generally, [70] uses conditionally invertible networks to provide a generic transfer between latent spaces of diverse domains. Different from VQ-VAEs, VQGANs [23, 103] employ a first stage with an adversarial and perceptual objective to scale autoregressive transformers to larger images. However, the high compression rates required for feasible ARM training, which introduces billions of trainable parameters [23, 66], limit the overall performance of such approaches and less compression comes at the price of high computational cost [23, 66]. Our work prevents such trade-offs, as our proposed *LDMs* scale more gently to higher dimensional latent spaces due to their convolutional backbone. Thus, we are free to choose the level of compression which optimally mediates between learning a powerful first stage, without leaving too much perceptual compression up to the generative diffusion model while guaranteeing high-fidelity reconstructions (see Fig. 1).

While approaches to jointly [93] or separately [80] learn an encoding/decoding model together with a score-based prior exist, the former still require a difficult weighting between reconstruction and generative capabilities [11] and are outperformed by our approach (Sec. 4), and the latter focus on highly structured images such as human faces.

3. Method

To lower the computational demands of training diffusion models towards high-resolution image synthesis, we observe that although diffusion models allow to ignore perceptually irrelevant details by undersampling the corresponding loss terms [30], they still require costly function evaluations in pixel space, which causes huge demands in computation time and energy resources.

We propose to circumvent this drawback by introducing an explicit separation of the compressive from the generative learning phase (see Fig. 2). To achieve this, we utilize an autoencoding model which learns a space that is perceptually equivalent to the image space, but offers significantly reduced computational complexity.

Such an approach offers several advantages: (i) By leaving the high-dimensional image space, we obtain DMs which are computationally much more efficient because sampling is performed on a low-dimensional space. (ii) We exploit the inductive bias of DMs inherited from their UNet architecture [71], which makes them particularly effective for data with spatial structure and therefore alleviates the need for aggressive, quality-reducing compression levels as required by previous approaches [23, 66]. (iii) Finally, we obtain general-purpose compression models whose latent space can be used to train multiple generative models and which can also be utilized for other downstream applications such as single-image CLIP-guided synthesis [25].

3.1. Perceptual Image Compression

Our perceptual compression model is based on previous work [23] and consists of an autoencoder trained by combination of a perceptual loss [106] and a patch-based [33] adversarial objective [20, 23, 103]. This ensures that the reconstructions are confined to the image manifold by enforcing local realism and avoids bluriness introduced by relying solely on pixel-space losses such as L_2 or L_1 objectives.

More precisely, given an image $x \in \mathbb{R}^{H \times W \times 3}$ in RGB space, the encoder $\mathcal{E}$ encodes x into a latent representa-